

[14X030] Introduction to Computational Finance

Exercise #10

May 5, 2026

General instructions

Each student is expected to upload on Moodle a **zip** file named as *firstname_lastname*, containing the followings:

- A report in **pdf** format to answer exercise questions.
- The **code** used to generate the results of the report.
- If preferred, submit a unique **Jupyter notebook** with both code and report explanations.

Deeper instructions

- **Report:**
 - Answers to the TP questions. Figures and numerical results are necessary but **strictly not** sufficient: provide your analysis/comments as well.
 - Example: if your results surprise you, explain why and explain what the expected result was.
 - **Pay attention** to the presentation: axis labels, legend, etc.
- **Code:**
 - Implementation questions + code used to produce the results and figures in the report.
 - Programming language of your choice.
 - Do not use ChatGPT, or any other LLM/AI-assisted tools while writing down your code. Try by yourself, otherwise it will be counted as failed.
 - I **strongly recommend Python/Julia**: I won't be able to assist you in the same way if you use a different language.

Deadline

Any questions to: [Lorenzo Bini](#).

Upload on Moodle due by : **May 11, 2026 at 11:59 pm**

The Greeks

Let an asset S with initial value S_0 at $t = 0$. We consider both a European option to buy this asset (call) and to sell it (put), both with maturity $T = 1$ (in years) and strike price $K = 120$. We assume a constant volatility $\sigma = 20\%$ over their lifespan, and a risk-free rate $r = 5\%$.

Plotting the greeks

Using the Black-Scholes formula to evaluate the call and the put price as a function of the initial asset price S_0 , plot the evolution of:

- the price of the call/put;
- the Δ of the call/put;
- the Γ of the call/put.

Comment on the meaning of these graphs.

Hedging a short call

Suppose the asset price is $S_0 = 100$, and we sell 1000 calls.

- Which position should we have to be Δ -neutral?
- What is the profit of this strategy, supposing that $S_{0+\epsilon} = 105$? $S_{0+\epsilon} = 95$?
- In the two previous cases, compare the quality of this hedging strategy with having a fully naked (no position) or a fully covered option (buying a quantity of 1000 of the underlying asset).

Hedging a short put

Suppose the asset price is $S_0 = 100$, and we sell 1000 puts.

- Which position should we have to be Δ -neutral?
- What is the profit of this strategy, supposing that $S_{0+\epsilon} = 105$? $S_{0+\epsilon} = 95$?
- In the two previous cases, compare the quality of this hedging strategy with having a fully naked or a fully covered option.